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A FIRST COURSE ON DIFFERENTIAL EQUATIONS WITH MODELING APPLICATION BY DENNIS G. ZILL (10e)

LIST OF RESULTS

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1 Introduction to Differential Equations

Theorem 1.2.1 (*Existence of a Unique Solution*) Let R be a rectangular region in the xy-plane defined by $a \leq x \leq b$, $c \leq y \leq d$ that contains the point (x_0, y_0) in its interior. If $f(x, y)$ and $\partial f/\partial y$ are continuous on R, then there exists some interval $I_0 : (x_0 - h, x_0 + h)$, $h > 0$, contained in $[a, b]$, and a unique function $y(x)$ defined on I_0, that is a solution of the initial-value problem.

2 First-Order Differential Equations

Theorem 2.4.1 (*Criterion for an Exact Differential*) Let $M(x, y)$ and $N(x, y)$ be continuous and have continuous partial derivatives on a rectangle R in the xy-plane. Then a necessary and sufficient condition that $M(x, y)dx + N(x, y)dy$ be an exact differential is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \tag{1}$$

4 Higher-Order Differential Equations

Theorem 4.1.1 (*Existence of a Unique Solution*) Let $a_n(x), a_{n-1}(x), \ldots, a_0(x)$ and $g(x)$ be continuous on an interval I and let $a_n(x) \neq 0$ for every x in I. If $x = x_0$ is any point in I, then a solution $y(x)$ of the initial-value problem exists on I and is unique.

Theorem 4.1.2 (*Superposition Principle—Homogeneous Equations*) Let $y_1(x), y_2, \ldots, y_k(x)$ be solutions of a homogeneous n-th order linear differential equation. Then the linear combination

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x), \tag{2}$$

where c_i, $i = 1, 2, \ldots, k$ are arbitrary constants, is also a solution of the equation.

Theorem 4.1.3 (*Criterion for Linearly Independent Solutions*) Let $y_1(x), y_2(x), \ldots, y_n(x)$ be n solutions of a homogeneous n-th order linear differential equation on an interval I. Then the set of solutions is linearly independent on I iff the Wronskian $W(y_1, y_2, \ldots, y_n) \neq 0$ for every x in I, where

$$W(y_1, y_2, \ldots, y_n) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & & & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}. \tag{3}$$

Theorem 4.1.4 (*Existence of a Fundamental Set*) There exists a fundamental set of solutions for the homogeneous n-th order linear differential equation on an interval I.

Theorem 4.1.5 (*General Solution—Homogeneous Equations*) Let $y_1(x), y_2(x), \ldots, y_n(x)$ be a fundamental set of solutions of a homogeneous n-th order linear differential equation on an interval I. Then the *general solution* of the equation on the interval is given by

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x), \tag{4}$$

where the c_i, $i = 1, 2, \ldots, n$ are arbitrary constants.

Theorem 4.1.6 (*General Solution—Nonhomogeneous Equations*) Let $y_p(x)$ be any particular solution of a nonhomogeneous n-th order linear differential equation on an interval I and let $y_1, y_2, \ldots, y_n$ be a fundamental set of solutions of the associated homogeneous equation on the same interval. Then the *general solution* of the nonhomogeneous equation on the interval is given by

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x) + y_p(x), \tag{5}$$

where the c_i, $i = 1, 2, \ldots, n$ are arbitrary constants.

Theorem 4.1.7 (*Superposition Principle—Nonhomogeneous Equations*) Let $y_{p1}, y_{p2}, \ldots, y_{pk}$ be k particular solutions of a nonhomogeneous n-th order linear differential equation on an interval I corresponding, in turn, to k distinct functions $g_1(x), g_2(x), \ldots, g_k(x)$. That is, suppose y_{pi} denotes a particular solution of the corresponding differential equation

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_0(x)y = g_i(x), \quad i = 1, 2, \ldots, k. \tag{6}$$

then

$$y_p(x) = y_{p1}(x) + y_{p2}(x) + \cdots + y_{pk}(x) \tag{7}$$

is a particular solution of

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_0(x)y = g_1(x) + g_2(x) + \cdots + g_k(x). \tag{8}$$

Theorem 4.2.1 (*Reduction of Order*) Let $y_1(x)$ be a solution of the homogeneous n-th order linear differential equation on an interval I and that $y_1(x) \neq 0$ for all x in I. Then

$$y_2(x) = y_1(x) \int \frac{e^{-\int P(x)dx}}{y_1^2(x)} dx \tag{9}$$

is a second solution of the equation on the interval.

7 The Laplace Transform

Definition 7.1.1 (*Laplace Transform*) Let f be a function defined on $[0, \infty)$. Then the function F defined by

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt \tag{10}$$

is said to be the *Laplace transform* of f. The domain of F is the set of all s for which the improper integral converges.

Theorem 7.1.1 (*Transforms of Some Basic Functions*)

(a) $\mathcal{L}\{1\} = \dfrac{1}{s}$

(b) $\mathcal{L}\{t^n\} = \dfrac{n!}{s^{n+1}}$ for $n = 0, 1, 2, \ldots$

(c) $\mathcal{L}\{e^{at}\} = \dfrac{1}{s-a}$

(d) $\mathcal{L}\{\sin(\omega t)\} = \dfrac{\omega}{s^2+\omega^2}$

(e) $\mathcal{L}\{\cos(\omega t)\} = \dfrac{s}{s^2+\omega^2}$

(f) $\mathcal{L}\{\sinh(\omega t)\} = \dfrac{\omega}{s^2-\omega^2}$

(g) $\mathcal{L}\{\cosh(\omega t)\} = \dfrac{s}{s^2-\omega^2}$

Definition 7.1.2 (*Exponential Order*) A function f defined on $[0, \infty)$ is said to be of *exponential order* as $t \to \infty$ if there exist constants c, $M > 0$, and $T > 0$ such that $|f(t)| \leq Me^{ct}$ for all $t > T$. If f is of exponential order, then the Laplace transform $\mathcal{L}\{f(t)\} \leq Me^{ct}$ for all $t > T$.

Theorem 7.1.2 (*Sufficient Conditions for Existence*) If f is piecewise continuous on $[0, \infty)$ and of exponential order as $t \to \infty$, then the Laplace transform $\mathcal{L}\{f(t)\}$ exists for all $s > c$, where c is a constant such that $|f(t)| \leq Me^{ct}$ for all $t > T$.

Theorem 7.1.3 (*Behavior of $F(s)$ as $s \to \infty$*) If f is piecewise continuous on $[0, \infty)$ and of exponential order as $t \to \infty$ and $\mathcal{L}\{f(t)\} = F(s)$, then $\lim\limits_{s\to\infty} F(s) = 0$.

Theorem 7.2.1 (*Some Inverse Transforms*)

(a) $\mathcal{L}^{-1}\left\{\dfrac{1}{s}\right\} = 1$

(b) $\mathcal{L}^{-1}\left\{\dfrac{n!}{s^{n+1}}\right\} = t^n$ for $n = 0, 1, 2, \ldots$

(c) $\mathcal{L}^{-1}\left\{\dfrac{1}{s-a}\right\} = e^{at}$

(d) $\mathcal{L}^{-1}\left\{\dfrac{\omega}{s^2+\omega^2}\right\} = \sin(\omega t)$

(e) $\mathcal{L}^{-1}\left\{\dfrac{s}{s^2+\omega^2}\right\} = \cos(\omega t)$

(f) $\mathcal{L}^{-1}\left\{\dfrac{\omega}{s^2-\omega^2}\right\} = \sinh(\omega t)$

(g) $\mathcal{L}^{-1}\left\{\dfrac{s}{s^2-\omega^2}\right\} = cosh(\omega t)$

Theorem 7.2.2 (*Transform of a Derivative*) Suppose f, f', $\ldots$, f^{n-1} are continuous on $[0, \infty)$, and f, f', $\ldots$, f^{n-1} are of exponential order as $t \to \infty$. Then for $s > c$,

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \ldots - f^{(n-1)}(0) \tag{11}$$

where $F(s) = \mathcal{L}\{f(t)\}$.

Theorem 7.3.1 (*First Translation Theorem*) If $\mathcal{L}\{f(t)\} = F(s)$ exists for $s > c$ and a is any real rumber, then

$$\mathcal{L}\{e^{at} f(t)\} = F(s - a) \tag{12}$$

for $s > c + a$.

Theorem 7.3.2 (*Second Translation Theorem*) If $\mathcal{L}\{f(t)\} = F(s)$ exists for $s > c \geq 0$, and a is a positive constant, then

$$\mathcal{L}\{f(t-a)\,\mathcal{U}(t-a)\} = e^{-as} F(s) \tag{13}$$

for $s > c$.

Theorem 7.4.1 (*Derivatives of Transforms*) If the function f is piecewise continuous on $[0, \infty)$, of exponential order and $\mathcal{L}\{f(t)\} = F(s)$, then for $n = 1, 2, 3, \ldots$ and $s > c$,

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s). \tag{14}$$

Theorem 7.4.2 (*Convolution Theorem*) If the functions f and g are piecewise continuous on $[0, \infty)$, of exponential order and $\mathcal{L}\{f(t)\} = F(s)$, $\mathcal{L}\{g(t)\} = G(s)$, then for $s > c$,

$$\mathcal{L}\{f * g\} = \mathcal{L}\{f(t)\}\mathcal{L}\{g(t)\} = F(s)G(s). \tag{15}$$

Theorem 7.4.3 (*Transform of a Periodic Function*) If f is piecewise continuous $[0, \infty)$, of exponential order, and periodic with period T, then

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt. \tag{16}$$

Theorem 7.5.1 (*Transform of the Dirac Delta Function*) For $t_0 > 0$,

$$\mathcal{L}\{\delta(t - t_0)\} = e^{-st_0}. \tag{17}$$

8 Systems of Linear Differential Equations

Definition 8.1.1 (*Solution Vector*) A **solution vector** on an interval I is any column matrix

$$\mathbf{X} = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix} \tag{18}$$

whose entries are differentiable functions satisfying the system $\mathbf{X}' = \mathbf{AX} + \mathbf{F}$ on the interval.

Theorem 8.1.1 (*Existence of a Unique Solution*) Let the entries of the matrices $\mathbf{A}(t)$ and $\mathbf{F}(t)$ be functions continuous on a common interval I that contains the point t_0. Then there exists a unique solution of the initial-value problem $\mathbf{X}' = \mathbf{A}(t)\mathbf{X} + \mathbf{F}(t)$, $\mathbf{X}(t_0) = \mathbf{X_0}$ on the interval.

Theorem 8.1.2 (*Superposition Principle*) Let $\mathbf{X}_1$, $\mathbf{X}_2$, ..., $\mathbf{X}_k$ be a set of solution vectors of the homogeneous system $\mathbf{X}' = \mathbf{AX}$ on an interval I. Then the linear combination

$$\mathbf{X} = c_1\mathbf{X}_1 + c_2\mathbf{X}_2 + \cdots + c_k\mathbf{X}_k, \tag{19}$$

where c_i, $i = 1, 2, \ldots, k$ are arbitrary constants, is also a solution of the system on the interval.

Theorem 8.1.3 (*Criterion for Linearly Independent Solutions*) Let

$$\mathbf{X}_1 = \begin{pmatrix} x_{11} \\ x21 \\ \vdots \\ x_{n1} \end{pmatrix}, \quad \mathbf{X}_2 = \begin{pmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{pmatrix}, \quad \ldots, \quad \mathbf{X}_n = \begin{pmatrix} x_{1n} \\ x_{2n} \\ \vdots \\ x_{nn} \end{pmatrix} \tag{20}$$

be n solution vectors of an homogeneous system on an interval I. Then the set of solution vectors is linearly independent on I iff the **Wronskian**

$$W(\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n) = \begin{vmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & & & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{vmatrix} \neq 0 \tag{21}$$

for every t in the interval.

Theorem 8.1.4 (*Existence of a Fundamental Set*) There exists a fundamental set of solutions for the homogeneous system $\mathbf{X}' = \mathbf{AX}$ on an interval I.

Theorem 8.1.5 (*General Solution—Homogeneous Systems*) Let $\mathbf{X}_1$, $\mathbf{X}_2$, ..., $\mathbf{X}_n$ be a fundamental set of solutions of a homogeneous system on an interval I. Then the **general solution** of the system on the interval is

$$\mathbf{X} = c_1\mathbf{X}_1 + c_2\mathbf{X}_2 + \cdots + c_n\mathbf{X}_n, \tag{22}$$

where the c_i, $i = 1, 2, \ldots, n$ are arbitrary constants.

Theorem 8.1.6 (*General Solution—Nonhomogeneous Systems*) Let $\mathbf{X}_p$ be a given solution of a nonhomogeneous system on an interval I and let

$$\mathbf{X}_c = c_1\mathbf{X}_1 + c_2\mathbf{X}_2 + \cdots + c_n\mathbf{X}_n \tag{23}$$

denote the general solution on the same interval of the associated homogeneous system. Then the **general solution** of the nonhomogeneous system on the interval is

$$\mathbf{X} = \mathbf{X}_c + \mathbf{X}_p. \tag{24}$$

The general solution $\mathbf{X}_c$ of the associated homogeneous system is called the **complementary function** of the nonhomogeneous system.

Theorem 8.2.1 (*Distinct Real Eigenvalues: General Solution—Homogeneous Systems*) Let λ_1, λ_2, ..., λ_n be n distinct real eigenvalues of the coefficient matrix $\mathbf{A}$ of the homogeneous system $\mathbf{X}' = \mathbf{AX}$ and let $\mathbf{K}_1$, $\mathbf{K}_2$, ..., $\mathbf{K}_n$ be the corresponding eigenvectors. Then the **general solution** of the system on the interval $(-\infty, \infty)$ is given by

$$\mathbf{X} = c_1\mathbf{K}_1e^{\lambda_1 t} + c_2\mathbf{K}_2e^{\lambda_2 t} + \cdots + c_n\mathbf{K}_ne^{\lambda_n t}. \tag{25}$$

Theorem 8.2.2 (*Solutions Corresponding to a Complex Eigenvalue*) Let A be the coefficient matrix having real entries of the homogeneous system $\mathbf{X}' = \mathbf{AX}$ and let $\mathbf{K}_1$ be an eigenvector corresponding to the complex eigenvalue $\lambda_1 = \alpha + i\beta$, α and β real numbers. Then

$$\mathbf{K}_1e^{\lambda_1 t} \quad \text{and} \quad \overline{\mathbf{K}_1}e^{\overline{\lambda_1}t} \tag{26}$$

are solutions of the system, where $\overline{\mathbf{K}_1}$ is the complex conjugate of $\mathbf{K}_1$ and $\overline{\lambda_1} = \alpha - i\beta$.

Theorem 8.2.3 (*Real Solutions Corresponding to a Complex Eigenvalue*) Let $\lambda_1 = \alpha + i\beta$ be a complex eigenvalue of the coefficient matrix $\mathbf{A}$ of the homogeneous system $\mathbf{X}' = \mathbf{AX}$ and let $\mathbf{B}_1$ and $\mathbf{B}_2$ denote the column vectors defined as $\mathbf{B}_1 = \frac{1}{2}\left(\mathbf{K}_1 + \overline{\mathbf{K}_1}\right)$ and $\mathbf{B}_2 = \frac{i}{2}\left(-\mathbf{K}_1 + \overline{\mathbf{K}_1}\right)$. Then

$$\mathbf{X}_1 = \mathbf{B}_1e^{\alpha t}\cos(\beta t) + \mathbf{B}_2e^{\alpha t}\sin(\beta t) \tag{27}$$

$$\mathbf{X}_2 = -\mathbf{B}_2e^{\alpha t}\cos(\beta t) + \mathbf{B}_1e^{\alpha t}\sin(\beta t) \tag{28}$$

are linearly independent solutions of the system $\mathbf{X}' = \mathbf{AX}$ on the interval $(-\infty, \infty)$.

The following is an introduction to difference equations, based on *An Introduction to Difference Equations* (3e), by S. Elaydi.

1 Dynamics of First-Order Difference Equations

A linear *homogeneous* first-order difference equation is given by

$$x(n+1) = a(n)x(n), \quad x(n_0) = x_0, \quad n \geq n_0 \geq 0, \tag{29}$$

and the associated *nonhomogeneous* equation is

$$y(n+1) = a(n)y(n) + g(n), \quad y(n_0) = y_0, \quad n \geq n_0 \geq 0, \tag{30}$$

where in both cases $a(n) \neq 0$, and $a(n)$ and $g(n)$ are real-valued functions defined for $n \geq n_0$.

The solution to (29) can be obtained by iteration and mathematical induction and is given by

$$x(n) = \left[\prod_{i=n_0}^{n-1} a(i)\right] x_0 \tag{31}$$

For the nonhomogeneous case, the unique solution is

$$y(n) = \left[\prod_{i=n_0}^{n-1} a(i)\right] y_0 + \sum_{r=n_0}^{n-1} \left[\prod_{i=r+1}^{n-1} a(i)\right] g(r). \tag{32}$$

Definition 1.5 A point x^* in the domain of f is said to be an *equilibrium point* of a difference equation

$$x(n+1) = f(x(n)) \tag{33}$$

if it is a fixed point of f, i.e., $f(x^*) = x^*$.

Definition 1.6 Let x be a point in the domain of f. If there exists a positive integer r and an equilibrium point x^* of (33) such that $f^r(x) = x^*$, $f^{r-1} \neq x^*$, then x is an *eventually equilibrium (fixed) point.*

Definition 1.8

(a) The equilibrium point x^* of (33) is *stable* if given $\epsilon > 0$ there exists $\delta > 0$ such that $|x_0 - x^*| < \delta$ implies $|f^n(x_0) - x^*| < \epsilon$ for all $n > 0$. If x^* is not stable, then it is called *unstable.*

(b) The point x^* is said to be *attracting* if there exists $\eta > 0$ such that

$$|x(0) - x^*| < \eta \implies \lim_{n\to\infty} x(n) = x^*.$$

If $\eta = \infty$, x^* is called a *global attractor* or *globally attracting.*

(c) The point x^* is an *asymptotically stable equilibrium point* if it is stable and attracting. If $\eta = \infty$, x^* is said to be *globally asymptotically stable.*